Reg. No. | | | | | | | | | | |

[17CS44]

FOURTH SEMESTER B.TECH EXAMINATION, JUNE – 2019

MICROCONTROLLER AND EMBEDDED SYSTEMS

Time:3 Hours **Maximum Marks:70**

Instructions:

1. Instructions: Answer any **FIVE** full questions.Choosing at least **TWO** questions from each Part.
2. Any missing data may be suitably assumed

PART - A

1.	**a.**	With a neat block diagram ,explain internal architecture of 8051.	**10**
	b.	Explain PSW register.	**04**
2.	**a.**	Perform the following operation using appropriate addressing modes in 8051: **i.** Read a data from code memory location 8030h and store the data in internal RAM 70h. Use offset address as 10h. **ii.** Exchange two bytes from internal data memory location 50h and 51h to external RAM Location 9000h and 9001h using indexed addressing mode.	**06**
	b.	Give the bit size and Data range details for the widely used C Data types of 8051.	**04**
	c.	Write an 8051 C program to toggle all the bits of P1 continuously.	**04**
3.	**a.**	Write the steps required for programming 8051 to transfer and receive data serially.	**08**
	b.	Write an ALP to transfer letter "Y" serially at 9600 baud rate continuously.	**06**
4.	**a.**	Explain the importance of TI and RI flag.	**06**
	b.	Define interrupt, and mention the difference between interrupt and polling method and also write the steps in executing interrupt.	**08**

PART-B

5.	**a.**	Explain DAC interface with diagram and also write a C program to generate staircase waveform.	**06**
	b.	Show the interfacing circuit and functional pins of LCD.	**08**
6.	**a.**	Discuss the categories of embedded systems.	**06**
	b.	Explain embedded system architecture.	**08**
7.	**a.**	Explain the layers required to construct embedded systems.	**07**
	b.	Explain the process of translation of code on target.	**07**
8.	**a.**	Explain RFID system components.	**07**
	b.	Explain Characteristics and types of RFID tags.	**07**

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